

PluralTree: Problem Formulation and Methodology

1 Problem formulation

1.1 Overton coverage

Let x be a question on which a population holds several distinct viewpoints. Let $W(x) = \{w_1, \dots, w_m\}$ be the set of viewpoint clusters that real respondents to x hold. A language model M produces a single answer $y = M(x)$. We say y *covers* cluster w when a majority of the people holding w judge that y represents their view.

Definition 1 (Overton coverage).

$$\text{Cov}(y, x) = \frac{1}{|W(x)|} \sum_{w \in W(x)} \mathbf{1}[w \in R(y)], \quad R(y) = \{w \in W(x) : \bar{\rho}_w(y) \geq 4\},$$

where $\bar{\rho}_w(y)$ is the mean 1–5 representation rating of y by holders of w . The benchmark score is the mean of Cov over questions.

Coverage is *unweighted* over clusters: a minority viewpoint counts the same as a majority one. This is what distinguishes the construct from distributional pluralism, which would weight by prevalence.

1.2 The single-answer constraint and its decomposition

The task requires *one* answer. Any system producing y must therefore (i) identify the viewpoints, (ii) get them into the generator, and (iii) fit them into one response. We name these stages and treat them separately throughout, because they fail separately:

stage	object	diagnostic
RETRIEVE	a set of viewpoint contrasts $F(x)$	union gain over a no-retrieval answer
DELIVER	a conditioning channel for $F(x)$	coverage with vs. without $F(x)$ at fixed M
COMPRESS	one y from several drafts	coverage of y vs. union of its drafts

A gain in the final score can originate at any stage; a null can be produced by any stage destroying what an earlier one supplied. Every experiment below is designed to vary one stage with the other two held fixed.

2 The viewpoint graph

2.1 Structure

The graph is a rooted tree $T = (V, E)$ built from population survey data. Each survey question q is a node; beneath it, one node per demographic *axis* (age, party, religion, ...); beneath each axis, one *opinion leaf* ℓ per subgroup, carrying that subgroup’s empirical answer distribution

$$p_\ell \in \Delta^{m_q-1}, \quad \sum_o p_\ell(o) = 1,$$

over the m_q options of q . Questions are grouped under topic and subtopic nodes. In the primary graph (Pew American Trends Panel; 1,492 questions, 58 subpopulations) the topic and subtopic layers are *induced* by k -means over sentence embeddings of question text; they are not native to the survey. This matters for the geometry ablation (§7).

2.2 Embedding

Every node $v \in V$ is embedded as $\phi(v) \in \mathbb{B}_c^d$, the Poincaré ball of curvature $-c$, by a tree-recurrent encoder with gated injection of node text features. The embedding is trained on link prediction over the tree’s edges and reported with structural-fidelity metrics: rank correlation of depth with radius, of tree distance with geodesic distance, ancestor-AUC, and a sibling-separation ratio. Setting $c = 0$ recovers Euclidean space with the same encoder; this is the geometry ablation.

3 Retrieval: the divergence scout

Given x , the scout returns a small set of *forks*. A fork is a triple (a, u, v) : an anchor node a and two of its children u, v whose subtrees disagree.

Relevance. Let $e(x)$ be a sentence embedding of x and $t(v)$ the same embedding of node v ’s text. Define $\text{rel}(v) = \cos(e(x), t(v))$.

anchors. $A(x)$ is the set of the n_a most relevant internal nodes after lexical matching on entities named in x .

Gated point clouds. For a child u of an anchor, let $S(u) = \{ \phi(z) : z \in \text{subtree}(u), \text{rel}(z) \geq \tau \}$, truncated to at most n_s nodes, with a floor of $n_{\min}$ so that a branch is never emptied. Each point carries mass $\mu_u(z) \propto \exp(\text{rel}(z)/T)$.

Divergence. For siblings u, v under a ,

$$\mathcal{W}(u, v) = \min_{\gamma \in \Pi(\mu_u, \mu_v)} \sum_{z \in S(u)} \sum_{z' \in S(v)} \gamma(z, z') d_c(\phi(z), \phi(z')),$$

the optimal-transport distance between the two subtrees’ embedding clouds under the hyperbolic ground metric d_c . Under $c = 0$, d_c is Euclidean.

Scoring and selection. Each candidate pair is scored

$$s(u, v) = \max(\bar{\text{rel}}(u, v), 0)^\alpha \cdot \mathcal{W}(u, v), \quad \bar{\text{rel}}(u, v) = \frac{1}{2}(\bar{\text{rel}}(S(u)) + \bar{\text{rel}}(S(v))),$$

and the scout returns $F(x) = \text{top-}k_{(u,v)}s(u, v)$ over all anchors. Defaults: $\tau = 0.25$, $\alpha = 1$, $k = 5$, $n_a = 4$, $n_s = 32$, $T = 0.1$.

Remark 1. *Selection is the only place the divergence enters. Everything downstream sees $F(x)$ as a set of node triples. The selection ablation therefore replaces $s(u, v)$ by a deterministic hash-based uniform rank over the same candidate pool, so that anchors, gating, fork count and rendering are byte- identical and only which pairs survive differs.*

4 Delivery: rendering and injection

Each fork is rendered to text by a fixed function ρ : a header naming the anchor with its divergence and relevance, the two poles’ option texts, and up to n_p *driver pairs* (z, z') from the transport plan γ with the largest $\gamma(z, z') d_c$. A *full-spectrum* variant instead renders every opinion leaf under the anchor, ordered by projection onto the pole-to-pole axis $p_v - p_u$ in distribution space.

The context $\rho(F(x))$ is placed in the prompt with an instruction I . The model generates a reasoning span, then an answer span; only the answer is scored. This is the delivery channel that the CAD experiment (§6) shows to be harmful.

5 Compression: multi-draft merge

Let $\mathcal{D} = \{(I_i, \beta_i, \sigma_i)\}_{i=1}^{n_d}$ be draft specifications: an instruction, whether forks are injected, and whether the full-spectrum render is used. The default is

$$\mathcal{D} = \{(\text{plain}, 0, 0), (\text{scout}, 1, 0), (\text{distributional}, 1, 1)\}.$$

Drafts are generated independently,

$$y_i = M(I_i, x, \beta_i \cdot \rho_{\sigma_i}(F(x))),$$

and merged by a single further call $\hat{y} = M(I_{\text{merge}}, x, \{y_i\})$ instructed to keep one paragraph per distinct position and drop nothing.

Guard. The merge is accepted only if it did not compress its inputs. With $|\cdot|_w$ word count and $\pi(\cdot)$ the number of paragraphs of at least $d_{\min}$ words,

$$|\hat{y}|_w \geq \lambda_\ell \max_i |y_i|_w, \quad \pi(\hat{y}) \geq \lambda_\pi \max_i \pi(y_i),$$

with $\lambda_\ell = \lambda_\pi = 1$. An optional *semantic* check dedupes the drafts’ deep units by embedding, then requires that a fraction $\geq \lambda_r$ of them have a unit of $\hat{y}$ within cosine θ_r ($\lambda_r = 0.8$, $\theta_r = 0.55$). On failure the drafts are concatenated under neutral headings, which is lossless in content by construction.

Remark 2. *The concatenation fallback is lossless in text, not in judged coverage: the judge rates a response as a whole, and a good draft appended to two poor ones is rated as a mostly-poor response. This distinction is what the 7B scale result exposed.*

Variants. PERSONA_MERGE keeps n_d and the single merge call but conditions drafts $2..n_d$ on individual subgroup vantage points ℓ chosen by spreading along the pole-to-pole axis (poles first, then widest gap), each told to commit to that view. MERGE_V2_RAND replaces $F(x)$ with structurally matched forks from an unrelated anchor (depth-matched, rendered-length-matched, one per real fork).

6 Decoding-level interventions

Context-aware decoding. With ℓ_{ctx} the next-token logits under the fork-injected prompt and $\ell_{\emptyset}$ under the same instruction with the fork block removed,

$$\ell_{\text{cad}}(\alpha) = (1 + \alpha) \ell_{\text{ctx}} - \alpha \ell_{\emptyset} = \ell_{\emptyset} + (1 + \alpha) (\ell_{\text{ctx}} - \ell_{\emptyset}).$$

α scales the context’s contribution: $\alpha > 0$ amplifies, $\alpha \in (-1, 0)$ interpolates toward the no-context model, $\alpha = -1$ recovers it exactly. In probability space this is the geometric blend $p \propto p_{\text{ctx}}^{1+\alpha} p_{\emptyset}^{-\alpha}$.

Graph-guided G2. Three streams decode in lockstep: a base stream z seeing only x , and two guide streams seeing prior answers, one told to diverge and one to replicate. With H the entropy of $\text{softmax}(z)$,

$$\ell = z + \alpha_t (z^+ - z^-), \quad \alpha_t = \theta \cdot \mathbf{1}[H(z) \geq \beta].$$

The graph variant replaces the generic “diverge” instruction with a named subgroup vantage point from $F(x)$. The forks never enter the base stream; they contribute a direction, not text.

7 Ablations and controls

Each ablation changes exactly one term of the pipeline above.

name	stage	what varies	what is held fixed
geometry $c \in \{0.5, 0\}$	retrieve	ground metric d_c , encoder curvature	training config, seed, $\mathcal{D}$, judge
DIVRAND	retrieve	rank key $s \rightarrow$ uniform hash	anchors, gate, k , ρ , merge
FLAT	retrieve	no tree: leaves ranked by rel alone, gated at τ	k , driver count, ρ , merge
RAND	retrieve	fork <i>content</i> (unrelated anchor)	fork count, depth, length, merge
CAD α	deliver	context influence	prompt, sampler, RNG stream
scale 7B vs 72B	deliver	generator (and judge, when one GPU serves both)	questions, $\mathcal{D}$
SEM guard	compress	semantic check on/off	drafts, structural guard

Manipulation checks. An ablation whose arms do not actually differ yields a plausible table that means nothing. Every retrieved condition therefore records per row its selection mode, fork count, mean $\mathcal{W}$ over selected forks, mean relevance, and the anchor pool examined. DIVRAND is valid only if its fork count equals the reference’s and its mean $\mathcal{W}$ is strictly lower (the reference takes the argmax). RAND is valid only if fork counts match and relevance is lower. FLAT is valid only if it abstains on the same questions the reference abstains on. These are asserted by a job that gates every full run.

8 Evaluation protocol

Score. OvertonScore as in Definition 1, with an LLM judge predicting each participant’s rating; $\bar{\rho}_w$ is the mean predicted rating over the participants in w . *The judge’s validity is unresolved.* The benchmark’s reported agreement ($\rho = 0.88$, a rank correlation over 8 reference models) belongs to the benchmark authors’ judge. Reproducing that check with this project’s judge (Qwen2.5-72B-Instruct-AWQ) gave aggregate $\rho = 0.167$ (an earlier run with too few participants per question gave 0.357), within-participant $\rho = 0.059$, and a generosity gap of $P(\hat{r} \geq 4) = 0.747$ against $P(r \geq 4) = 0.650$ for humans. The achievable ceiling — human split-half agreement at the same level — has not been measured, so it is not known whether 0.167 is poor or near that limit. Per-cluster agreement is likewise unestablished. Every coverage number below inherits this.

Paired comparison. For conditions c_1, c_2 on shared questions Q , the estimand is

$$\bar{\delta} = \frac{1}{|Q|} \sum_{x \in Q} [\text{Cov}(y_{c_1}(x), x) - \text{Cov}(y_{c_2}(x), x)],$$

with a 95% interval and two-sided p from a 10,000-draw bootstrap over *questions*. Because resampling is over questions, additional rollouts per question reduce within-question noise only; between-question heterogeneity sets the floor on interval width.

Difference of differences. For an ablation with arms g_1, g_2 , $\Delta\Delta = \bar{\delta}_{g_1} - \bar{\delta}_{g_2}$ bootstrapped over the same questions. Two arms that are individually significant and non-significant against baseline have *not* thereby been shown to differ.

Union, oracle, gain. With $R_c(x)$ the covered set under condition c :

$$\begin{aligned} \text{union}(x) &= \left| \bigcup_c R_c(x) \right| / |W(x)|, \\ \text{oracle}(x) &= \max_c |R_c(x)| / |W(x)|, \\ \text{gain} &= \overline{\text{union}} - \max_c \overline{|R_c| / |W|}. \end{aligned}$$

Union $\geq$ oracle $\geq$ best single, by construction. Gain measures what an arm *adds* to another, not how it scores; two arms can be equal alone and differ several-fold in gain. It is a ceiling on any merge, not a score.

Noise floor. The baseline condition uses no retrieval and cannot depend on any retrieval-side change. Its spread across otherwise identical runs is therefore a direct estimate of run-to-run noise, and any delta of that size is not distinguishable from zero.

9 What is and is not established

The calibration of every claim above — what is established, what is ruled out as a driver, what remains untested, and the judge-validity caveat that bounds all of it — is kept in a companion document, `established_claims.tex`, so that this file stays a specification of the method. Numbers there are dated and carry their confidence intervals; the running record as results land is `findings.2026_09_15.md`.